ChE 2O04 Tutorial 1 – Introduction to Microsoft Excel

All undergraduate students currently enrolled at McMaster qualify for the free Office 365 subscription, which includes free downloads of the Office applications for Windows and Mac. Go to: https://office365.mcmaster.ca/microsoft-365-for-students-start-here/

- Ensure that Microsoft Excel is installed on your computer along with the following add-ins:
  a. Analysis ToolPak
  b. Solver

(Very) Basic Fundamentals:

- Prior to continuing, make sure that you understand the following:

i. Entering information into a cell - an equal sign at the beginning tells Excel to do a calculation. For example, try the following:

3+6	vs.	=3+6

ii. Using formulas in cells, for example:

a. Square roots: $\sqrt{4} \rightarrow$ =SQRT(4)

b. Pi: π = 3.1415926535… → =PI()

c. Logarithmic functions: ln(5) → =ln(5); $\log_{10}(5) \rightarrow$ =log(5)

iii. *Absolute* vs. *relative* cell references:

Enter the following into a worksheet:

	A	B	C	D	E
1	3	8	14		
2	7	12	21		
3	14	17	28		
4					
5					

a. Type "=A1" into cell E5, then copy (using shortcut Ctrl+C) the formula from cell E5 and paste (using shortcut Ctrl+V) it into cells F5, G5, E6, F6, G6, E7, F7, G7.

b. Repeat step **a.** but now using the formula "=A1" in cell E5; notice the big difference between that from part **a.**? **c.** Repeat step **a.** but now using the formula "=A$1" in cell E5.

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iv. Trigonometric functions expect the arguments to be in *radians, not degrees*.

π (radians) = 180 (degrees)

sin(90°) → =sin(90*PI()/180)
=sin(radians(90))

v. Common syntax errors:

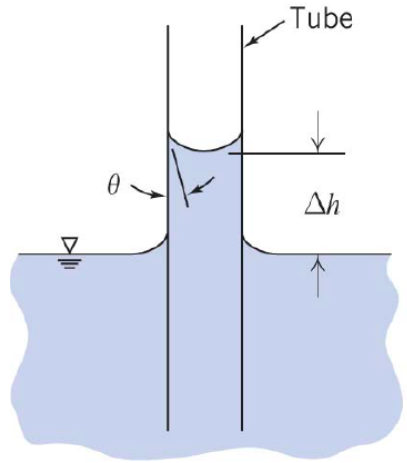
#DIV/0!	Division by zero, or something *close* to zero
#NAME?	Formula contains an undefined variable or function name
#NUM!	Numeric overflow
#REF!	Invalid cell reference
#VALUE!	Invalid argument type
#######	Not an error; column is too narrow to display number

vi. Solving equations numerically:

a. Goal Seek — Data > What-If Analysis > Goal Seek…
b. Solver — Data > Solver

Tutorial Activities:

1. Later on in the course we will see that due to surface tension effects, when a capillary tube with a rather narrow internal diameter is lowered into a liquid bath that the water will "climb" up the inside of the capillary to a height Δh above the liquid surface. This is shown schematically in the figure to the right; you can also see a demonstration of this event at around the 9:30 mark in this online video. The height of the fluid inside the capillary can be calculated according to:



$$\Delta h = \frac{2\sigma \cos\theta}{\rho g r_{capillary}}$$

where σ is the surface tension of the liquid (N/m)
θ is the contact angle between the liquid and the capillary tube
ρ is the density of the liquid (kg/m^3)
g is the gravitational constant (9.81 m/s^2)
$r_{capillary}$ is the radius of the capillary tube (m)

Calculate Δh for when $\sigma = 0.07$ N/m, $\theta = 28°$, $\rho = 998$ kg/m^3, $g = 9.81$ m/s^2, and $r_{capillary} = 0.5$ mm.

2. The Darcy-Weisbach Equation for specific energy loss h_l due to friction between the moving fluid in a pipe is:

$$h_l = f\,\frac{L}{D}\,\frac{V_{avg}^2}{2}$$

where f is the friction factor (a dimensionless number), L is the pipe length (m), D is the pipe diameter (m), and V_{avg} is the *average* fluid velocity (m/s).

a. Prove that h_l has units of length^2/time^2; **then show that those units are equivalent to energy/ mass.**

b. Using Excel, calculate V_{avg} for a fluid flow rate of 0.04 m^3/s and a pipe diameter of 10 cm.
Hint: To solve this, you should NOT use the Darcy-Weisbach equation.

c. Using Excel, calculate h_l for the values given in part **b**, friction factor of 0.02, and pipe length of 3 m.

d. Using Excel, create a nice figure of h_l versus flow rate (from 0.01 to 1 m^3/s) using the pipe diameter, friction factor, and pipe length values from parts **b** and **c**.

3. The friction factor f in the Darcy-Weisbach Equation can be estimated using the Colebrook Equation:

$$\frac{1}{\sqrt{f}} + 2.0\log_{10}\left[\frac{\varepsilon}{3.7} + \frac{2.51}{Re\sqrt{f}}\right] = 0$$

where ε is the relative roughness of the pipe (dimensionless) and Re is the Reynolds number (dimensionless). Using the Solver tool in Excel, calculate f for a relative roughness of 0.004 and a Reynolds number of 200,000.
Hint: For your initial guess of f use 0.01.